

```

#handling inconsistency data
#time series operations -- time series data is data collected or recorded over a period of time.
#Creating a Date Column
import pandas as pd
data = {
    "Date": [
        "2025-01-01",
        "2025-01-02",
        "2025-01-03",
        "2025-01-04",
        "2025-01-05"
    ],
    "Sales": [5000, 6500, 4200, 7000, 5500]}
df = pd.DataFrame(data)
print(df)

print("-----")

print(df.dtypes)

print("-----")

#Converting Column to Datetime

df["Date"] = pd.to_datetime(df["Date"])
print(df.dtypes)

print("-----")

#4.pd.to_datetime()
#pd.to_datetime() converts values into Pandas datetime format.

date = pd.to_datetime("2025-08-22")
print(date)

print("-----")

#5. Generating Synthetic Time-Series Data----pd.date_range()

#6. pd.date_range() --- This generates dates from January 1 to January 10.

dates = pd.date_range(
    start="2025-01-01",
    end="2025-01-10"
)
print(dates)

print("-----")

#7. Generating a Fixed Number of Dates----- Instead of specifying an end date, we can specify the number of periods.

dates = pd.date_range(
    start="2025-01-01",
    periods=10
)
print(dates)

print("-----")

#8. Generating Data with Frequency
#We can specify how frequently dates should be generated.

#Daily -- D - day
dates = pd.date_range(start="2025-01-01",periods=10,freq="D")
print(dates)

print("-----")

#Hourly - h
dates = pd.date_range(start="2025-01-01",periods=10,freq="h")
print(dates)

print("-----")

#minute - min
dates = pd.date_range(start="2025-01-01",periods=10,
    freq="min")
print(dates)

print("-----")

```

```
#Monthly -- MS - monthstart, ME - monthend
dates = pd.date_range(start="2025-01-01",periods=12,freq="ME")
print(dates)
```

```
      Date  Sales
0  2025-01-01  5000
1  2025-01-02  6500
2  2025-01-03  4200
3  2025-01-04  7000
4  2025-01-05  5500
```

```
Date      object
Sales     int64
dtype: object
```

```
Date      datetime64[ns]
Sales     int64
dtype: object
```

```
2025-08-22 00:00:00
```

```
DatetimeIndex(['2025-01-01', '2025-01-02', '2025-01-03', '2025-01-04',
               '2025-01-05', '2025-01-06', '2025-01-07', '2025-01-08',
               '2025-01-09', '2025-01-10'],
              dtype='datetime64[ns]', freq='D')
```

```
DatetimeIndex(['2025-01-01', '2025-01-02', '2025-01-03', '2025-01-04',
               '2025-01-05', '2025-01-06', '2025-01-07', '2025-01-08',
               '2025-01-09', '2025-01-10'],
              dtype='datetime64[ns]', freq='D')
```

```
DatetimeIndex(['2025-01-01', '2025-01-02', '2025-01-03', '2025-01-04',
               '2025-01-05', '2025-01-06', '2025-01-07', '2025-01-08',
               '2025-01-09', '2025-01-10'],
              dtype='datetime64[ns]', freq='D')
```

```
DatetimeIndex(['2025-01-01 00:00:00', '2025-01-01 01:00:00',
               '2025-01-01 02:00:00', '2025-01-01 03:00:00',
               '2025-01-01 04:00:00', '2025-01-01 05:00:00',
               '2025-01-01 06:00:00', '2025-01-01 07:00:00',
               '2025-01-01 08:00:00', '2025-01-01 09:00:00'],
              dtype='datetime64[ns]', freq='h')
```

```
DatetimeIndex(['2025-01-01 00:00:00', '2025-01-01 00:01:00',
               '2025-01-01 00:02:00', '2025-01-01 00:03:00',
               '2025-01-01 00:04:00', '2025-01-01 00:05:00',
               '2025-01-01 00:06:00', '2025-01-01 00:07:00',
               '2025-01-01 00:08:00', '2025-01-01 00:09:00'],
              dtype='datetime64[ns]', freq='min')
```

```
DatetimeIndex(['2025-01-31', '2025-02-28', '2025-03-31', '2025-04-30',
               '2025-05-31', '2025-06-30', '2025-07-31', '2025-08-31',
               '2025-09-30', '2025-10-31', '2025-11-30', '2025-12-31'],
              dtype='datetime64[ns]', freq='ME')
```

```
#questions
```

```
#Q1.Generate 5 years of monthly dates starting from 2020-01-01 using periods. Create a DataFrame with a randomly generated
```

```
import pandas as pd
import numpy as np
dates = pd.date_range(start="2020-01-01", periods=60, freq="MS")
```

```
df = pd.DataFrame({
    "Date": dates,
    "Sales": np.random.randint(1000, 10000, size=60)
})
```

```
print(df)
```

```
      Date  Sales
0  2020-01-01  7785
1  2020-02-01  1739
2  2020-03-01  2436
3  2020-04-01  9348
4  2020-05-01  3432
5  2020-06-01  5056
6  2020-07-01  5209
7  2020-08-01  1191
8  2020-09-01  3764
9  2020-10-01  3592
10 2020-11-01  4155
11 2020-12-01  6089
12 2021-01-01  9751
```

```
13 2021-02-01 8499
14 2021-03-01 2682
15 2021-04-01 8487
16 2021-05-01 4918
17 2021-06-01 5692
18 2021-07-01 4864
19 2021-08-01 4531
20 2021-09-01 1958
21 2021-10-01 1711
22 2021-11-01 1823
23 2021-12-01 7738
24 2022-01-01 5292
25 2022-02-01 2595
26 2022-03-01 9227
27 2022-04-01 1275
28 2022-05-01 3862
29 2022-06-01 6230
30 2022-07-01 2059
31 2022-08-01 2478
32 2022-09-01 2257
33 2022-10-01 1425
34 2022-11-01 8362
35 2022-12-01 3812
36 2023-01-01 8980
37 2023-02-01 1684
38 2023-03-01 2285
39 2023-04-01 4877
40 2023-05-01 6017
41 2023-06-01 9689
42 2023-07-01 2932
43 2023-08-01 2609
44 2023-09-01 1287
45 2023-10-01 8426
46 2023-11-01 2834
47 2023-12-01 6918
48 2024-01-01 7322
49 2024-02-01 7023
50 2024-03-01 2657
51 2024-04-01 3441
52 2024-05-01 7304
53 2024-06-01 3776
54 2024-07-01 3750
55 2024-08-01 5985
56 2024-09-01 6708
```

```
#Q2.Generate 365 days of daily dates starting from 2024-01-01 using periods. Generate random daily sales between ₹1,000 and
dates = pd.date_range(start="2024-01-01", periods=365, freq="D")
```

```
df = pd.DataFrame({
    "Date": dates,
    "Sales": np.random.randint(1000, 10000, size=365)
})

print(df)
```

	Date	Sales
0	2024-01-01	6912
1	2024-01-02	9325
2	2024-01-03	8834
3	2024-01-04	2256
4	2024-01-05	8332
..	...	...
360	2024-12-26	3727
361	2024-12-27	9186
362	2024-12-28	3607
363	2024-12-29	2387
364	2024-12-30	1735

```
[365 rows x 2 columns]
```

```
#Q3.Generate 24 months of monthly sales data starting from 2023-01-01 using periods. Create Date and Sales columns.
```

```
dates = pd.date_range(start="2023-01-01", periods=24, freq="MS")
```

```
df = pd.DataFrame({
    "Date": dates,
    "Sales": np.random.randint(1000, 10001, size=24)
})

print(df)
```

	Date	Sales
0	2023-01-01	7927
1	2023-02-01	1112
2	2023-03-01	3284
3	2023-04-01	3494
4	2023-05-01	8016
5	2023-06-01	1797

```
6 2023-07-01 5273
7 2023-08-01 9403
8 2023-09-01 2738
9 2023-10-01 5462
10 2023-11-01 7435
11 2023-12-01 7090
12 2024-01-01 9475
13 2024-02-01 7839
14 2024-03-01 7480
15 2024-04-01 9012
16 2024-05-01 4141
17 2024-06-01 9201
18 2024-07-01 1743
19 2024-08-01 9423
20 2024-09-01 1992
21 2024-10-01 6971
22 2024-11-01 5124
23 2024-12-01 5353
```

```
#Q4.Generate 30 days of hourly sales data starting from 2024-01-01 00:00:00 using periods. Create random sales values for €
dates = pd.date_range(start = "2024-01-01 00:00:00", periods = 720, freq = "h")
df = pd.DataFrame({
    "Date" : dates,
    "Sales" : np.random.randint(1000, 10001, size=720)
})
print(df)
```

	Date	Sales
0	2024-01-01 00:00:00	4135
1	2024-01-01 01:00:00	4521
2	2024-01-01 02:00:00	5244
3	2024-01-01 03:00:00	2138
4	2024-01-01 04:00:00	4916
..	...	...
715	2024-01-30 19:00:00	2451
716	2024-01-30 20:00:00	6003
717	2024-01-30 21:00:00	1230
718	2024-01-30 22:00:00	2773
719	2024-01-30 23:00:00	2696

[720 rows x 2 columns]

```
#Q5.Generate 5 years of monthly sales data from 2020 using periods. After creating the DataFrame, extract the Year and Month
#12 * 5 = 60
dates = pd.date_range(start = "2020-01-01", periods = 60, freq = "MS")
df = pd.DataFrame({
    "Date": dates,
    "Sales": np.random.randint(1000, 10001, size=60)
})

print(df)
print("-----")
#extract
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
print(df)

print("-----")
yearly_sales = df.groupby("Year")["Sales"].sum()

print(yearly_sales)
```

	Date	Sales
0	2020-01-01	1802
1	2020-02-01	8391
2	2020-03-01	8975
3	2020-04-01	4740
4	2020-05-01	5119
5	2020-06-01	3714
6	2020-07-01	7364
7	2020-08-01	8376
8	2020-09-01	3035
9	2020-10-01	4713
10	2020-11-01	5134
11	2020-12-01	3958
12	2021-01-01	8140
13	2021-02-01	5519
14	2021-03-01	8229
15	2021-04-01	1093
16	2021-05-01	4337
17	2021-06-01	6540
18	2021-07-01	3215
19	2021-08-01	1484
20	2021-09-01	1412
21	2021-10-01	9659

```

22 2021-11-01 6231
23 2021-12-01 4948
24 2022-01-01 4044
25 2022-02-01 3472
26 2022-03-01 2419
27 2022-04-01 2644
28 2022-05-01 9806
29 2022-06-01 6441
30 2022-07-01 1005
31 2022-08-01 3550
32 2022-09-01 7799
33 2022-10-01 7630
34 2022-11-01 6589
35 2022-12-01 7673
36 2023-01-01 9850
37 2023-02-01 6955
38 2023-03-01 1212
39 2023-04-01 4154
40 2023-05-01 1643
41 2023-06-01 5204
42 2023-07-01 6137
43 2023-08-01 6034
44 2023-09-01 2455
45 2023-10-01 5737
46 2023-11-01 6085
47 2023-12-01 5681
48 2024-01-01 5647
49 2024-02-01 2914
50 2024-03-01 1892
51 2024-04-01 9410
52 2024-05-01 8059
53 2024-06-01 2397
54 2024-07-01 6056
55 2024-08-01 4340
56 2024-09-01 4901

```

Start coding or [generate](#) with AI.

```

import pandas as pd

data = {
    "OrderID": [
        1001, 1002, 1003, 1004, 1005,
        1006, 1007, 1008, 1009, 1010,
        1011, 1012, 1013, 1014, 1015
    ],
    "OrderDateTime": [
        "2024-01-05 09:15:20",
        "2024-02-10 14:30:45",
        "2024-03-15 18:20:10",
        "2024-04-20 11:45:30",
        "2024-05-25 20:10:55",
        "2024-06-12 08:35:15",
        "2024-07-18 16:25:40",
        "2024-08-22 19:50:25",
        "2024-09-05 10:20:35",
        "2024-10-14 13:40:50",
        "2024-11-20 17:15:10",
        "2024-12-25 21:30:45",
        "2025-01-10 09:45:20",
        "2025-02-15 15:10:35",
        "2025-03-20 18:55:50"
    ],
    "Product": [
        "Laptop", "Mobile", "Tablet", "Headphones", "Keyboard",
        "Mouse", "Monitor", "Laptop", "Mobile", "Tablet",
        "Headphones", "Laptop", "Keyboard", "Monitor", "Mobile"
    ],
    "Sales": [
        55000, 25000, 30000, 5000, 3500,
        1500, 18000, 60000, 27000, 32000,
        6500, 58000, 4000, 20000, 29000
    ]
}

df = pd.DataFrame(data)

print(df)

# 1. Convert to datetime
df["OrderDateTime"] = pd.to_datetime(df["OrderDateTime"])

```

```
print(df)
```

	OrderID	OrderDateTime	Product	Sales
0	1001	2024-01-05 09:15:20	Laptop	55000
1	1002	2024-02-10 14:30:45	Mobile	25000
2	1003	2024-03-15 18:20:10	Tablet	30000
3	1004	2024-04-20 11:45:30	Headphones	5000
4	1005	2024-05-25 20:10:55	Keyboard	3500
5	1006	2024-06-12 08:35:15	Mouse	1500
6	1007	2024-07-18 16:25:40	Monitor	18000
7	1008	2024-08-22 19:50:25	Laptop	60000
8	1009	2024-09-05 10:20:35	Mobile	27000
9	1010	2024-10-14 13:40:50	Tablet	32000
10	1011	2024-11-20 17:15:10	Headphones	6500
11	1012	2024-12-25 21:30:45	Laptop	58000
12	1013	2025-01-10 09:45:20	Keyboard	4000
13	1014	2025-02-15 15:10:35	Monitor	20000
14	1015	2025-03-20 18:55:50	Mobile	29000

	OrderID	OrderDateTime	Product	Sales
0	1001	2024-01-05 09:15:20	Laptop	55000
1	1002	2024-02-10 14:30:45	Mobile	25000
2	1003	2024-03-15 18:20:10	Tablet	30000
3	1004	2024-04-20 11:45:30	Headphones	5000
4	1005	2024-05-25 20:10:55	Keyboard	3500
5	1006	2024-06-12 08:35:15	Mouse	1500
6	1007	2024-07-18 16:25:40	Monitor	18000
7	1008	2024-08-22 19:50:25	Laptop	60000
8	1009	2024-09-05 10:20:35	Mobile	27000
9	1010	2024-10-14 13:40:50	Tablet	32000
10	1011	2024-11-20 17:15:10	Headphones	6500
11	1012	2024-12-25 21:30:45	Laptop	58000
12	1013	2025-01-10 09:45:20	Keyboard	4000
13	1014	2025-02-15 15:10:35	Monitor	20000
14	1015	2025-03-20 18:55:50	Mobile	29000

```
# 1. Convert to datetime
df["OrderDateTime"] = pd.to_datetime(df["OrderDateTime"])
print(df.dtypes)

print("-----")

# 2. Extract Year
df["Year"] = df["OrderDateTime"].dt.year
print(df)

print("-----")

# 3. Extract Month
df["Month"] = df["OrderDateTime"].dt.month
print(df)

print("-----")

# 4. Extract Day
df["Day"] = df["OrderDateTime"].dt.day
print(df)

print("-----")

# 5. Extract Hour, Minute, Second
df["Hour"] = df["OrderDateTime"].dt.hour
df["Minute"] = df["OrderDateTime"].dt.minute
df["Second"] = df["OrderDateTime"].dt.second
print(df)

print("-----")
```

```
OrderID          int64
OrderDateTime    datetime64[ns]
Product          object
Sales            int64
Year             int32
Month            int32
Day              int32
Hour             int32
Minute           int32
Second           int32
dtype: object
```

	OrderID	OrderDateTime	Product	Sales	Year	Month	Day	Hour	\
0	1001	2024-01-05 09:15:20	Laptop	55000	2024	1	5	9	
1	1002	2024-02-10 14:30:45	Mobile	25000	2024	2	10	14	
2	1003	2024-03-15 18:20:10	Tablet	30000	2024	3	15	18	

3	1004	2024-04-20	11:45:30	Headphones	5000	2024	4	20	11
4	1005	2024-05-25	20:10:55	Keyboard	3500	2024	5	25	20
5	1006	2024-06-12	08:35:15	Mouse	1500	2024	6	12	8
6	1007	2024-07-18	16:25:40	Monitor	18000	2024	7	18	16
7	1008	2024-08-22	19:50:25	Laptop	60000	2024	8	22	19
8	1009	2024-09-05	10:20:35	Mobile	27000	2024	9	5	10
9	1010	2024-10-14	13:40:50	Tablet	32000	2024	10	14	13
10	1011	2024-11-20	17:15:10	Headphones	6500	2024	11	20	17
11	1012	2024-12-25	21:30:45	Laptop	58000	2024	12	25	21
12	1013	2025-01-10	09:45:20	Keyboard	4000	2025	1	10	9
13	1014	2025-02-15	15:10:35	Monitor	20000	2025	2	15	15
14	1015	2025-03-20	18:55:50	Mobile	29000	2025	3	20	18

	Minute	Second
0	15	20
1	30	45
2	20	10
3	45	30
4	10	55
5	35	15
6	25	40
7	50	25
8	20	35
9	40	50
10	15	10
11	30	45
12	45	20
13	10	35
14	55	50

	OrderID	OrderDateTime	Product	Sales	Year	Month	Day	Hour \
0	1001	2024-01-05 09:15:20	Laptop	55000	2024	1	5	9
1	1002	2024-02-10 14:30:45	Mobile	25000	2024	2	10	14
2	1003	2024-03-15 18:20:10	Tablet	30000	2024	3	15	18
3	1004	2024-04-20 11:45:30	Headphones	5000	2024	4	20	11
4	1005	2024-05-25 20:10:55	Keyboard	3500	2024	5	25	20
5	1006	2024-06-12 08:35:15	Mouse	1500	2024	6	12	8
6	1007	2024-07-18 16:25:40	Monitor	18000	2024	7	18	16
7	1008	2024-08-22 19:50:25	Laptop	60000	2024	8	22	19
8	1009	2024-09-05 10:20:35	Mobile	27000	2024	9	5	10
9	1010	2024-10-14 13:40:50	Tablet	32000	2024	10	14	13
10	1011	2024-11-20 17:15:10	Headphones	6500	2024	11	20	17

```
#simple apply()
import pandas as pd
data = {
    "EmpID" : [101,102,103,104,105],
    "Name" : ["Ravi","PRIYA","Arun","sneha","Kiran"],
    "City" : ["Hyderabad","Hyderabad","CHENNAI","Chennai","Bangalore"],
    "salary" : [30000,40000,3500,4500,32000],
    "Age" : [22,23,34,45,24]
}
df = pd.DataFrame(data)
print(df)
print("-----")

df["salary"] = df["salary"].apply(
    lambda x:x * 0.10
)
print(df)
```

	EmpID	Name	City	salary	Age
0	101	Ravi	Hyderabad	30000	22
1	102	PRIYA	Hyderabad	40000	23
2	103	Arun	CHENNAI	3500	34
3	104	sneha	Chennai	4500	45
4	105	Kiran	Bangalore	32000	24

	EmpID	Name	City	salary	Age
0	101	Ravi	Hyderabad	3000.0	22
1	102	PRIYA	Hyderabad	4000.0	23
2	103	Arun	CHENNAI	350.0	34
3	104	sneha	Chennai	450.0	45
4	105	Kiran	Bangalore	3200.0	24

```
#map-- to replace one value with another value
import pandas as pd
city_codes = {
    "Hyderbad" : "hyd",
    "bangalore" : "bngr",
    "Delhi" : "del"
}

df["city_codes"] = df["City"].map(city_codes)
print(df)
```

	EmpID	Name	City	salary	Age	city_codes
0	101	Ravi	Hyderabad	3000.0	22	NaN
1	102	PRIYA	Hyderabad	4000.0	23	NaN
2	103	Arun	CHENNAI	350.0	34	NaN
3	104	sneha	Chennai	450.0	45	NaN
4	105	Kiran	Bangalore	3200.0	24	NaN